Random Matrices

M1R Project

Dr Sheehan Olver

Overview

I Central Limit Theorem: Before digging in to random matrices we discuss a baby-version of universality coming from adding many random variables, using the Method of Moments to relate to the number of partitions of a set into pairs.

II Universality: Many families of random matrices have the same eigenvalue statistics which can be explained by adapting the Method of Moments to relate to the number of *non-crossing* partitions of a set into pairs.

III Free Probability: Adding or multiplying random matrices satisfy special rules of (non-commutative) free random variables. These can be explained again by the Method of Moments.

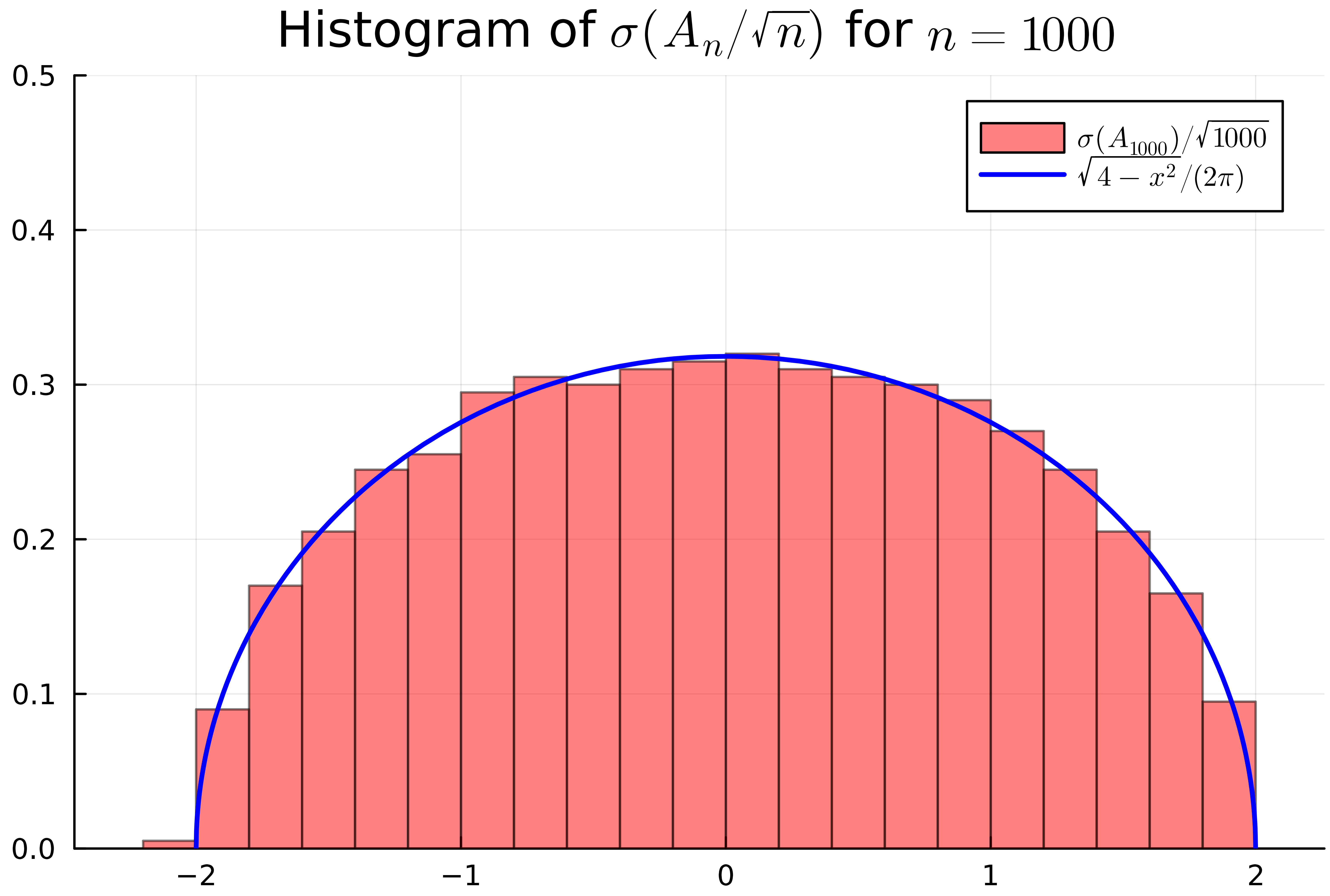
IV Randomised Linear Algebra: We finish with a nice application of using randomness to reduce linear algebra problems like linear regression to lower dimensional versions which are faster to solve.

Link to notes

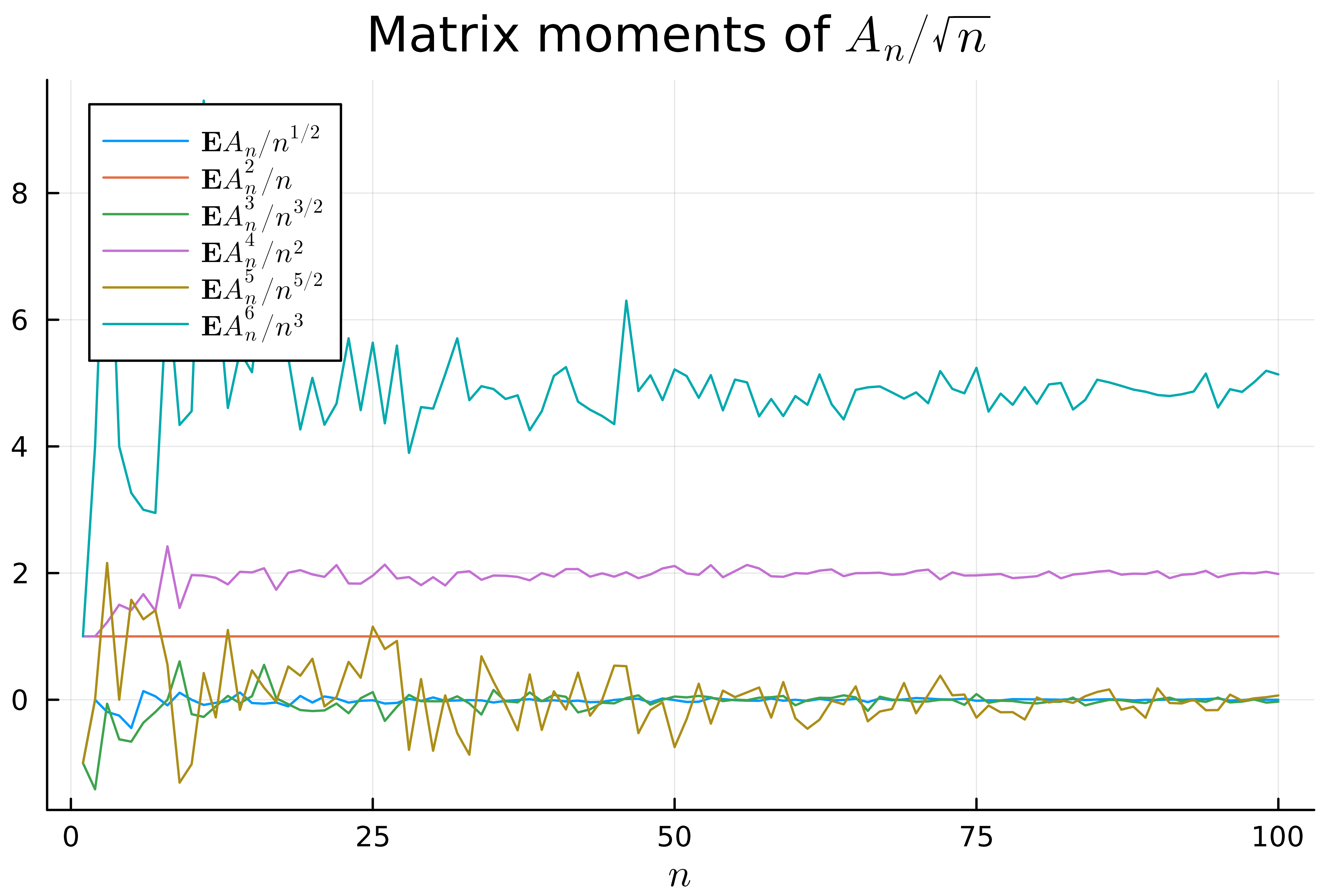
II Universality

II.1 Experimental observation of universality

Eigenvalues of random variables look like semicircle



Histogram of $\sigma(A_n/\sqrt{n})$ for $n = 1000$
$\sigma(A_{1000})/\sqrt{1000}$
$\sqrt{4-x^2}/(2\pi)$
0.0
0.1
0.2
0.3
0.4
0.5
−2
−1
0
1
2



Matrix moments of $A_n/\sqrt{n}$
$\mathbf{E}A_n/n^{1/2}$
$\mathbf{E}A_n^2/n$
$\mathbf{E}A_n^3/n^{3/2}$
$\mathbf{E}A_n^4/n^2$
$\mathbf{E}A_n^5/n^{5/2}$
$\mathbf{E}A_n^6/n^3$
0
2
4
6
8
0
25
50
75
100
n

II.2 Wigner ensembles and global statistics

How do we understand eigenvalue distributions?

Definition 2 (Wigner)**.** A random matrix $A_n \in \mathbb{R}^{n\times n}$ is a Wigner matrix if:

1. The entries a_{kj} for $j \geq k$ are all iid with mean 0, variance 1, and all moments are finite.
2. The matrix A_n is symmetric: $a_{kj} = a_{jk}$.

Proposition 1 (Cyclic property of trace)**.** *For any* $A \in \mathbb{R}^{n \times m}$ *and* $B \in \mathbb{R}^{m \times n}$ *we have*

$$\operatorname{Tr}[AB] = \operatorname{Tr}[BA]$$

Proposition 2 (Traces, powers and eigenvalues). *For any symmetric* $A \in \mathbb{R}^{n\times n}$ *with eigenvalues* $\lambda_1, \ldots, \lambda_n$ *we have*

$$\operatorname{Tr} A^k = \lambda_1^k + \ldots + \lambda_n^k.$$

II.3 Moments of the semicircle distribution

Expressible as Catalan numbers

Definition 3 (Catalan numbers).

$$C_m := \frac{1}{m+1}\binom{2m}{m} = \frac{(2m)!}{(m+1)!m!}.$$

Lemma 3 (Catalan recurrence).

$$C_{m+1} = \sum_{k=0}^{m} C_k C_{m-k}.$$

Lemma 4 (non-crossing pair partitions). *The number of partitions of* $2m$ *into non-crossing pairs equals* C_m*.*

Lemma 5 (semicircle moments).

$$\int_{-2}^{2} x^k \frac{\sqrt{4-x^2}}{2\pi} \mathrm{d}x = \begin{cases} 0 & \textit{if } k \textit{ is odd} \\ C_{k/2} & \textit{if } k \textit{ is even} \end{cases}$$

II.4 Proof of the semicircle law

Relate moments of sums to partitions

Write $$A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix}.$$

Proposition 3 (Trace of powers)**.** *For any k we have*

$$\operatorname{Tr} A^k = \sum_{i_1,\ldots,i_k=1}^{n} a_{i_1 i_2} a_{i_2 i_3} \cdots a_{i_k i_1}.$$

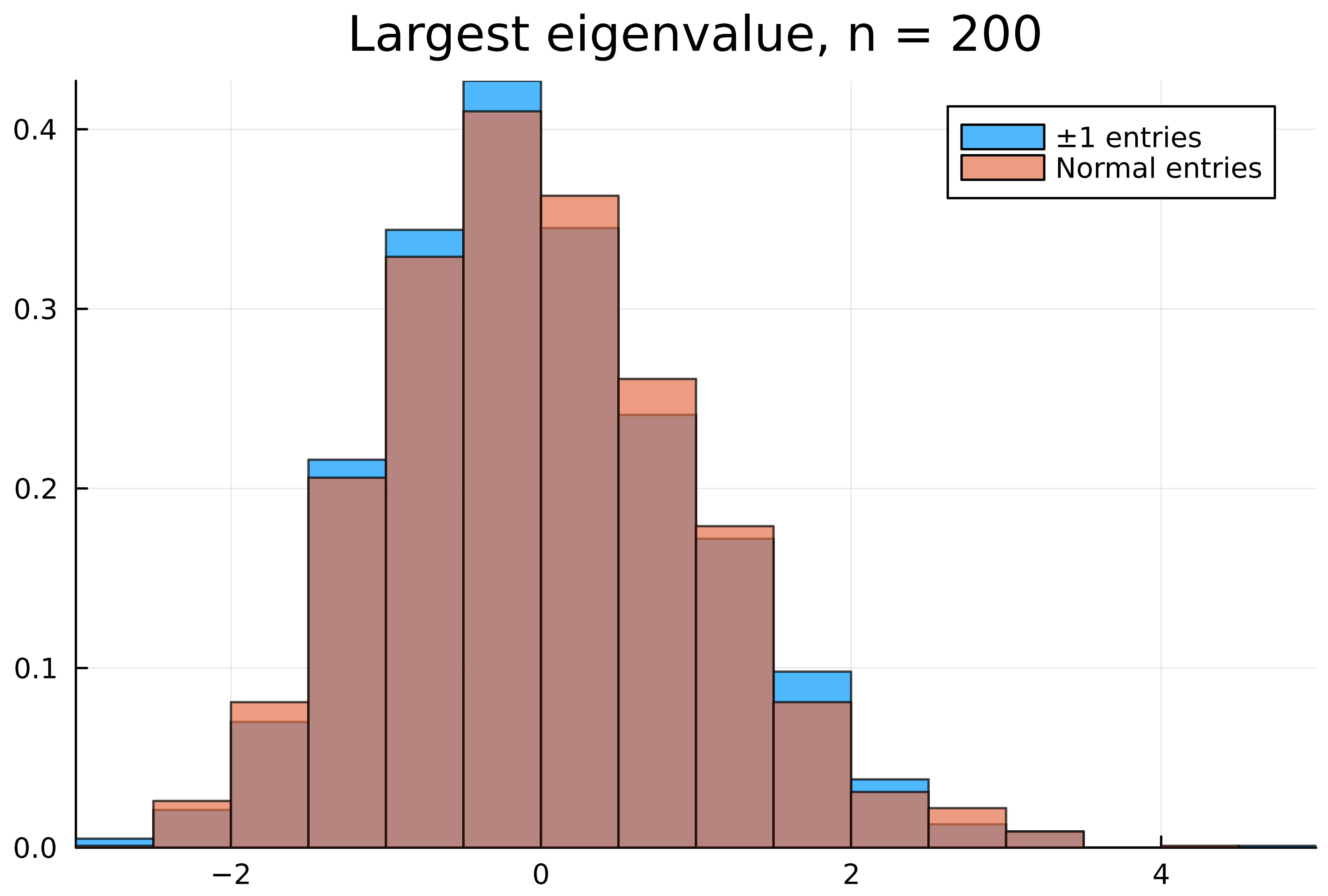
Theorem 2 (semicircle law)**.** *For any Wigner ensemble A_n we have*

$$\frac{\mathbf{E}A_n^k}{n^{k/2}} \xrightarrow[n\to\infty]{} \begin{cases} C_{k/2} & \text{if } k \text{ is even} \\ 0 & \text{if } k \text{ is odd} \end{cases}.$$

Partition	Term	Count
$\{1\},\{2\},\{3\},\{4\}$	$\mathbb{E}[a_{i_1 i_2} a_{i_2 i_3} a_{i_3 i_4} a_{i_4 i_1}] = 0$	$(n+1)n(n-1)(n-2) = n^4 + O(n^3)$
$\{1,2\},\{3\},\{4\}$	Not possible!	0
$\{1,3\},\{2\},\{4\}$	$\mathbb{E}[a_{i_1 i_2} a_{i_2 i_2} a_{i_2 i_1} a_{i_1 i_1}] = 0$	$n(n-1) = n^2 + O(n)$
$\{1,4\},\{2\},\{3\}$	Not possible!	0
$\{1\},\{2,3\},\{4\}$	Not possible!	0
$\{1\},\{2,4\},\{3\}$	$\mathbb{E}[a_{i_1 i_1} a_{i_1 i_3} a_{i_3 i_3} a_{i_3 i_1}] = 0$	$n(n-1) = n^2 + O(n)$
$\{1\},\{2\},\{3,4\}$	Not possible!	0
$\{1,2\},\{3,4\}$	$\mathbb{E}[a_{i_1 i_2} a_{i_2 i_1} a_{i_1 i_3} a_{i_3 i_1}] = \mu_2^2 = 1$	$n^2(n-1) = n^3 + O(n^2)$
$\{1,3\},\{2,4\}$	Not possible!	0
$\{1,4\},\{2,3\}$	$\mathbb{E}[a_{i_1 i_2} a_{i_2 i_3} a_{i_3 i_2} a_{i_2 i_1}] = \mu_2^2 = 1$	$n^2(n-1) = n^3 + O(n^2)$
$\{1,2,3\},\{4\}$	Not possible!	0
$\{1,2,4\},\{3\}$	Not possible!	0
$\{1,3,4\},\{2\}$	Not possible!	0
$\{1\},\{2,3,4\}$	Not possible!	0
$\{1,2,3,4\}$	$\mathbb{E}[a_{i_1 i_2} a_{i_2 i_1} a_{i_1 i_2} a_{i_2 i_1}] = \mu_4$	n^2

II.5 Local Eigenvalue Statistics

Largest eigenvalue also exhibit universality



Largest eigenvalue, n = 200
±1 entries
Normal entries
0.0
0.1
0.2
0.3
0.4
−2
0
2
4

II.6 Poster Projects

Ideas for posters related to universality

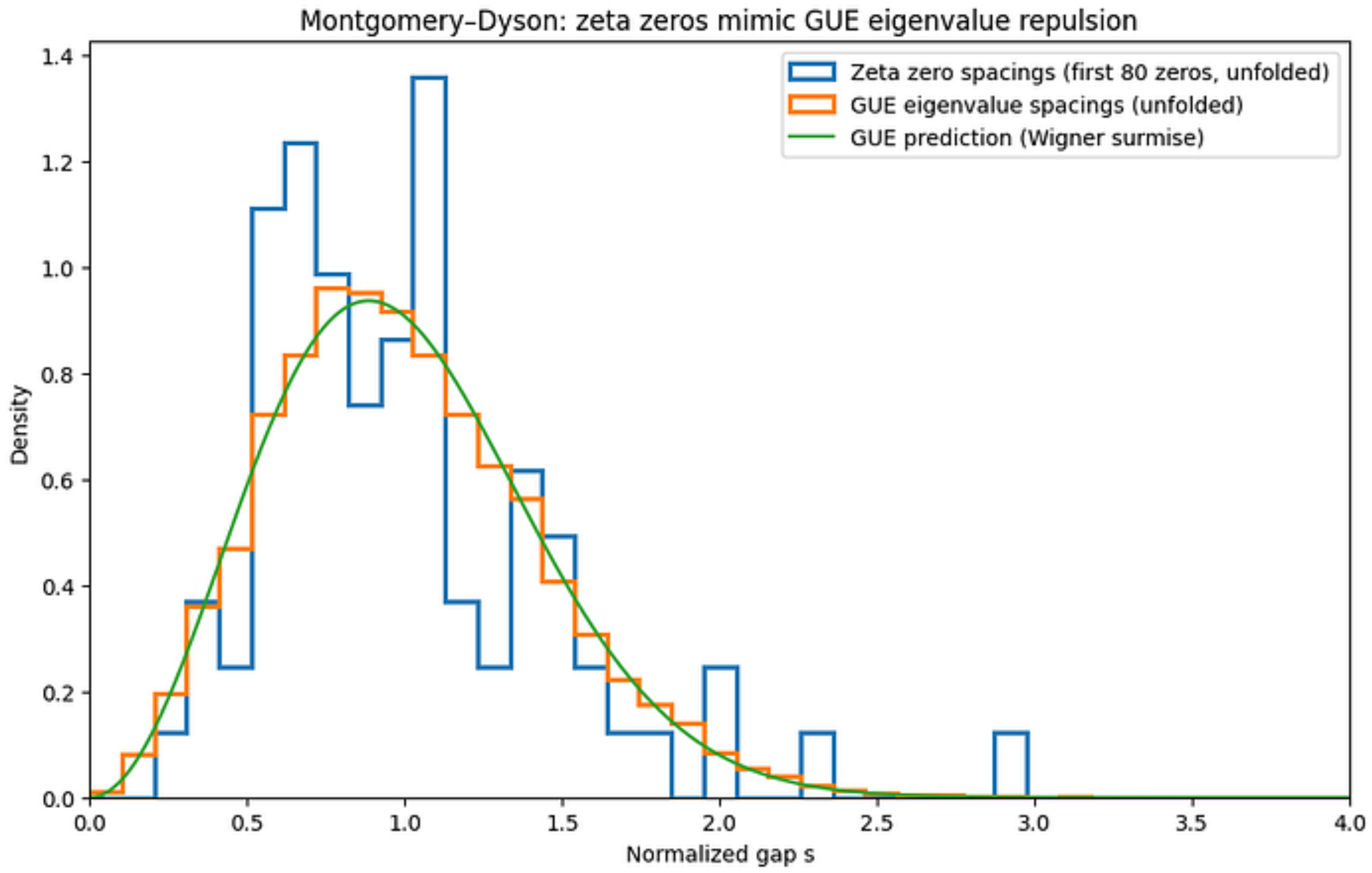
Detailed explanation of Catalan numbers

Local statistics in the bulk

Hermitian random matrices

Wishart Ensembles

Spacing of zeros of Riemann Zeta function



The Depth Behind the Simplest Rules: A Chronicle of Prime Numbers | by Daniel Rodríguez | The Quantastic Journal | Medium

Heavytailed random matrices

Random networks